American Economic Journal: Applied Economics 2024, 16(4): 1–33
https://doi.org/10.1257/app.20220621

The Curse of Plenty: The Green Revolution and the Rise in Chronic Disease†

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The rising rate of chronic disease is a leading driver of the global disease burden. Yet, its determinants are not fully understood. Exploiting the Green Revolution and its expansion in historically groundwater-rich Indian districts, we examine the unanticipated contribution of agricultural productivity growth to the rise in chronic, diet-related diseases. We find that areas with greater adoption of new staple varieties saw an increase in diabetes in men born after the introduction of high-yield crops. We find suggestive evidence that diet is an important mechanism, such as heterogeneous impacts with respect to dietary habits and increases in household calorie consumption. (*JEL* I12, O13, O15, O31, Q12, Q16)

Chronic diseases, such as diabetes and heart disease, are leading causes of mortality worldwide, accounting for 60 percent of all deaths in 2001 (WHO 2002). Previously considered "rich country diseases," chronic-disease-related deaths in poor countries have been growing rapidly (WHO 2016). Experts suggest that changes in diet are driving the global rise in chronic, metabolic conditions (Popkin 2001; Swinburn et al. 2011; Yach, Stuckler, and Brownell 2006), but the timing, nature, and determinants of these changes are not fully understood.

One possible contributing factor is technological advances in agriculture. Such progress aimed to address undernutrition, traditionally the focus of nutrition policy in developing countries. India, for example, saw rates of childhood stunting fall from 63 percent to 38 percent from 1989 to 2015 (World Bank 2023; International Institute for Population Studies Mumbai 2017a). However, growth in agricultural output can also give rise to diseases of plenty. In India, the rate of being overweight or obese among adults doubled from 10 percent to 20 percent from 2005 to 2015

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† Go to https://doi.org/10.1257/app.20220621 to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

(International Institute for Population Studies Mumbai 2017a),[1] and recent estimates suggest that 10 percent of adults have diabetes (World Bank 2023).[2] Many countries now struggle with a double burden of malnutrition, with high rates of undernutrition existing alongside growing disease burdens from metabolic conditions related to overconsumption.

This paper studies the extent to which agricultural productivity growth contributed to the rise in chronic diseases, by examining the Green Revolution in India. In 1966, the Indian government introduced high-yield varieties of rice and wheat across the country. The resulting growth in agricultural output allayed concerns about famine and food security, reduced food prices, and increased calorie availability (Evenson and Gollin 2003). Recent work on the Green Revolution by Gollin, Hansen, and Wingender (2021) highlights the importance of agricultural productivity growth to GDP in the developing world. In this paper, we focus on the unanticipated impact of exposure to the Green Revolution at young ages on chronic disease. In order to identify this effect, we exploit temporal variation in birth and historical geographic variation in the thickness of underground aquifers since the high-yield crops required timely irrigation. We first document that the adoption of high-yield varieties increased faster in districts with more abundant aquifers.

Next, using difference-in-differences (DID) and cohort event study models with two different household surveys, we find higher incidence of chronic diet-related diseases, driven by diabetes, among men born after 1966 in water-abundant districts. First, we harness the 2005 India Human Development Survey (IHDS), which includes self-reported measures of various diseases. Our IHDS sample includes men born between 1951 and 1981, allowing us to examine pre-existing trends and offer support for our identifying assumption that health outcomes would have trended similarly in the absence of the Green Revolution. Controlling for district-specific trends and state $\times$ birth-year fixed effects, we find robust evidence of an increase in the incidence of diabetes.

Second, we find corroborating evidence of an increase in diabetes using a second household survey conducted ten years later, the 2015–2016 National Family Health Survey (NFHS). An important advantage of the NFHS is that it collects biomarkers, including readings of glucose levels and blood pressure. A disadvantage is that the survey includes few cohorts of men born prior to the Green Revolution, thus limiting our pre-trend analysis. Nevertheless, using similar specifications, we find significant evidence of increases in glucose levels for men, but no evidence of impacts on other diseases. We estimate an increase in diabetes prevalence of about 4 percentage points, on a base of 26 percent among men older than 49 in 2015–2016. Diabetes prevalence varies significantly across states in India, ranging from 4.5–6 percent to more than 10.5 percent of adults (Tandon et al. 2018).[3]

[1] Overweight is defined as a body mass index (BMI) $\geq$ 25 kilograms per square meters ($\mathrm{kg/m^2}$), and obesity is defined as a BMI $\geq$ 30 $\mathrm{kg/m^2}$.

[2] Diabetes prevalence has been steadily increasing since 1971 and has increased fourfold between 1971 and 2014 in urban India (Mohan et al. 2018). In 2020, approximately 0.7 million Indians died of diabetes or complications arising from diabetes (International Diabetes Federation 2021).

[3] Interestingly, we do not find any significant effects on chronic disease for women. We discuss possible explanations in Section VB.

We consider various mechanisms for the impact of the Green Revolution on diabetes in men. Our strategy—comparing those born right after the distribution of high-yield varieties to those born before, in groundwater-rich districts—identifies the effect of exposure to the Green Revolution at very young ages (in utero and maybe before the age of 2, "early childhood" hereafter). Those born before the Green Revolution, our control group, would also be affected by the availability of high-yield crops, but at older ages. Our event study results reveal a sharp increase in diabetes risk for cohorts born in 1966–1968 relative to those born immediately before, and no evidence of differential pre-trends; that is, among cohorts born before 1966, diabetes prevalence appears unrelated to age at first exposure. We hypothesize that changes in nutrition during gestation and early childhood play an important role.

The study of the long-term consequences of in utero health originated with the Barker hypothesis that nutritional conditions in utero can program human genes for a greater tendency toward certain diseases and resistance to others (Barker 1990; Harding 2001; Kimm 2004). One theory is that poor nutrition in utero can lead to the development of a "thrifty phenotype," which allows the body to make the most out of calories available but can also lead to a greater incidence of metabolic syndrome in a nutrient-rich future. Our results run counter to the thrifty phenotype hypothesis since it would predict a greater risk of chronic disease for cohorts born *before* the Green Revolution, who would experience a greater mismatch between nutritional needs set in utero and calories available. At the same time, the in utero environment can lead to an increase in chronic disease through the composition of maternal diet; a low-protein maternal diet has been found to lead to high blood pressure and glucose intolerance in rats and mice (Dahri et al. 1991; Samuelsson et al. 2008; Nivoit et al. 2009; Stocker, Arch and Cawthorne 2005).

Thus, how in utero and early childhood exposure to high-yield varieties affects chronic disease depends on how dietary composition changed for pregnant women and young children after farmers adopted high-yield varieties. An immediate impact of the Green Revolution was to increase farm profits (Foster and Rosenzweig 1995). This increase in rural income along with greater agricultural output likely affected diet, but the change in the consumption of various macronutrients (fats, carbohydrates/sugar, protein) is ambiguous due to changes in the relative price of staples as well as whether staples are an inferior good. We do not have data on dietary intake for pregnant women and young children in the 1960s, but agricultural data indicate that the production of pulses, a primary source of protein in India, fell in groundwater-rich districts, as more agricultural land was allocated to staples.[4] Our results suggest that a shift toward a low-protein maternal diet outweighed the benefit of improved nutrition predicted by the thrifty phenotype hypothesis.

We provide suggestive evidence of the role played by a low-protein (maternal) diet. We show that the impact on diabetes varies with respect to an important, plausibly exogenous dietary habit. As highlighted in Atkin (2013, 2016), households in India differ in whether their primary staple is rice or wheat, and this dietary habit

[4] We assume that agricultural markets at the time of the Green Revolution were not integrated across districts, and so local production would have affected local prices more than prices in other districts.

has nutritional consequences. Relative to wheat, rice has half the amount of protein, more carbohydrates, less fat, and a higher glycemic index (Foster-Powell, Holt, and Brand-Miller 2002). The increase in diabetes is predominantly found among rice-eating households.[5,6]

We also consider alternative mechanisms and conclude that they are unlikely to explain our results. Most importantly, our preferred specification includes district fixed effects, state × birth-year fixed effects, and district-specific linear trends that should control for other potential drivers of chronic diseases. For example, our results are robust to other likely contributing factors to rising metabolic diseases in India, such as economic liberalization in the 1990s; recent trends in the consumption of sugary, processed, or prepared foods; and even trends in lifestyle, access to sedentary jobs, health care access, and the likelihood of diagnosis. Even if these trends were geographically correlated with groundwater availability, there is no reason to believe that they would affect men born after 1966 more than men before 1966, especially conditional on district-specific linear trends and state × birth-year fixed effects.[7]

That said, while our strategy identifies the effect of in utero and early childhood exposure, there could be other changes to the environment (besides nutrition) that might affect long-term health; for example, the Green Revolution ushered in modern agricultural practices such as the use of fertilizer. To explain our results, these agricultural practices would have to affect children in utero or in early childhood. Our results are robust to controlling for fertilizer use in the year of birth, and fertilizer exposure itself does not affect chronic diseases. We also find no evidence of an increase in other, non-diet-related chronic diseases such as cancer, asthma, and tuberculosis. Finally, we rule out composition effects that might occur if the Green Revolution affected which kinds of individuals survived to develop diabetes, discussed further in Section VIB.

This paper is inspired by the economics literature on the Barker hypothesis documenting long-term effects of income shocks in utero and during early childhood (see Almond and Currie 2011 and Almond, Currie, and Duque 2018 for comprehensive reviews). Papers exploiting changes in US welfare policies consistently find positive impacts on future health (Aizer et al. 2016; Hoynes, Schanzenbach, and Almond 2016). There is also a rich literature documenting the positive long-run impacts of improved in utero and perinatal nutrition in developing countries (McEniry and Palloni 2010; Almond and Mazumder 2011; Adhvaryu, Fenske, and Nyshadham 2019; Field, Robles, and Torero 2009; Maccini and Yang

[5] We first confirm that being a rice-eating household is not affected by the Green Revolution.

[6] Note that these nutritional differences are highly relevant for diabetes—the glycemic index measures how quickly food affects blood sugar levels—and, indeed, increased consumption of rice in India has been linked to increased risk of diabetes (Mohan et al. 2009). There is a recent debate on the dietary cause of obesity between the conventional energy balance model (calories in, calories out) and the newer carbohydrate-insulin model focusing on glycemic load (Ludwig and Ebbeling 2018; Hall, Guyenet, and Leibel 2018). Our results may have implications for this literature: the conventional model would predict that we would find impacts on other diet-related chronic diseases such as heart disease, while our stronger results for diabetes and rice-eating households could offer support for the carbohydrate-insulin model.

[7] Indian medical studies indicate that the prevalence of diabetes has been steadily rising since 1971 (Mohan et al. 2018, 2006) preceding the episode of economic liberalization in 1991.

2009; Shah and Steinberg 2017).[8] One particularly relevant paper, Hoddinott et al. (2008), finds positive long-term effects on economic productivity of a protein-rich supplement during early childhood, as compared to a caloric supplement with no protein, highlighting the importance of dietary composition.[9]

Our paper contributes to the literature in several ways. We study the long-term consequences of in utero and early childhood exposure to a positive and persistent economic shock in a developing country and, in contrast to related work, document a negative impact. Relative to work that studies extreme events, our study of the impact of greater access to nutrition is arguably more generalizable to the experience of economic growth in developing countries. In this way, we contribute to the literature on long-term impacts of agricultural productivity growth and help to understand the dual burden of malnutrition in developing countries. In a closely related paper, Hoynes, Schanzenbach, and Almond (2016) find that early childhood exposure to the Food Stamp Program in the United States led to a reduction in similar chronic diseases. While other differences between the United States and India could reconcile these results, one likely candidate is that Indian households changed their diet in a manner different from US households, possibly because of cultural dietary habits or because of changes in relative prices of various nutrients.[10] We highlight long-term consequences of in utero and early childhood exposure to calorie-rich, low-protein diets.

Finally, we contribute to recent research on the effect of the Green Revolution. Gollin, Hansen, and Wingender (2021) document the significant contribution of the Green Revolution to economic development, likely increasing well-being. However, there is also evidence of unanticipated adverse effects. Brainerd and Menon (2014) find that fertilizer use driven by the Green Revolution increased infant and child mortality, although Bharadwaj et al. (2020) and von der Goltz et al. (2020) find that the Green Revolution reduced infant mortality on net. In addition, Headey and Hoddinott (2016) document that the Green Revolution in Bangladesh improved child weight-for-height but not height-for-age, a nuanced effect of the Green Revolution on nutrition consistent with dietary substitution of carbohydrates for protein.[11] We complement this work and show that these health effects are not only limited to the short run but also matter for later life outcomes.

A full cost-benefit analysis of the Green Revolution is beyond the scope of this paper, but it is worth comparing the magnitude of the adverse effect we find to an important benefit documented in the literature. As discussed further in Section VIB, estimates from Bharadwaj et al. (2020) suggest a reduction in infant mortality of 0.2 percentage points in groundwater-rich districts relative to

[8] Yet another strand of this literature studies the long-run impact of negative shocks in childhood resulting in extreme deprivation (see Currie and Vogl 2013).

[9] In a follow-up study, Martorell et al. (2010) conclude that this protein-rich supplement did not increase the risk of heart disease. Since the two supplements differed in calories and protein content, it is not possible to isolate the two effects.

[10] Clay, Schmick, and Troesken (2019) also highlight the importance of crop diversification for nutrition, showing that the substitution away from cotton production in the American South in the early 1900s toward local food production reduced rates of niacin deficiency and pellagra despite also reducing income.

[11] Pingali (2012, 2015) and Gómez et al. (2013) also suggest that dietary diversity fell during the Green Revolution in India.

groundwater-poor districts. We find a 4 percentage point increase in the probability of having diabetes conditional on surviving to adulthood.[12] We use the concept of years of life lost to compare these magnitudes, which requires assumptions on survival to adulthood and the impact of diabetes on life expectancy. We estimate that the reduction in infant mortality would save 9–13 years of life per 100 live births.[13] At the same time, the increase in diabetes would result in approximately 7–12 years of life lost per 100 live births.[14] We interpret these estimates with caution given the many assumptions, but the similarity in magnitudes is striking.

The paper is organized as follows. Section I provides background on the Green Revolution in India necessary to understand the temporal and geographic variation we use, while Section II provides our theory of change. Section III describes the data, while Section IV details the estimation strategy. Section V describes the impact on adult health, and Section VI considers possible mechanisms. Section VII concludes.

I. The Green Revolution

A. *Timeline*

After obtaining independence from British rule in 1947, the Indian government was focused on strategies to foster a self-sufficient economy by investing in industrial development but also by addressing the persistent threat of food insecurity and the heavy reliance on food imports. Indian scientists experimented with irrigation infrastructure, chemical and organic fertilizers, and crop rotation practices, and while agricultural production improved, the rate of progress remained slow (Saha 2013).

Similar concerns about low levels of agricultural productivity and rising food insecurity globally motivated donor agencies such as the Ford Foundation and the Rockefeller Foundation to invest in agricultural research. The investment was fruitful: both the International Rice Research Institute (IRRI) in the Philippines and the International Centre for Maize and Wheat Improvement in Mexico were able to successfully conduct cross-breeding experiments in the early 1960s. These experiments were more successful for rice and wheat than other crops due to preexisting scientific knowledge rendering hybrid varieties of rice and wheat available earlier (Gollin, Hansen, and Wingender 2021).

In 1963, under a new agricultural strategy, the Indian Agriculture Research Institute sought the advice of Norman Borlaug, an agronomist affiliated with the Rockefeller Foundation who had been researching high-yield varieties of wheat in

[12] It is possible that this is a lower bound of the full effect of the Green Revolution on diabetes prevalence since we are only able to identify the effect of in utero and early childhood exposure; all cohorts were affected by the Green Revolution to some degree.

[13] We bound this estimate using life expectancy at birth in India in 1966 ($0.2 \times 46 = 9.2$) and life expectancy at birth in 2005 ($0.2 \times 64 = 12.8$) (World Bank 2023).

[14] Loukine et al. (2012) and Rosenquist and Fox (2016) provide estimates of the reduction in life expectancy of five to eight years from diabetes, although whether these estimates are applicable to India is unclear. Due to the challenge of identifying a causal effect, we use the conservative estimate of 5 years. We also use the probability of surviving to age 65 of 0.37 in 1966 ($4 \times 0.37 \times 5 = 7.4$) and 0.62 in 2005 ($4 \times 0.62 \times 5 = 12.4$) to bound this estimate (World Bank 2023). Data limitations prevent us from using the probability of surviving to age 40, which would be more consistent with our empirical analysis.

Mexico since 1944 (Saha 2013; Gollin, Hansen, and Wingender 2021). Borlaug determined that the hybrid wheat of semidwarf varieties would be most suitable for India, provided that large amounts of fertilizer and irrigation were available (Saha 2013). In 1965–1966, the Indian government introduced a semidwarf variety of wheat grown in Mexico and a similar high-yield variety of rice developed by the IRRI (Sen et al. 1974; Eliazer Nelson, Ravichandran, and Antony 2019).[15] Proving Borlaug right, these high-yield varieties succeeded in India. Food grain production increased dramatically (Gollin, Hansen, and Wingender 2021), and by 1975, around 62 percent of cereal production was attributed to high-yield varieties (Barker, Herdt, and Rose 1985). Sekhri (2014) plots the area of land farmed with high-yield varieties each year, demonstrating the dramatic rise starting in 1966 (see Figure A5 in the online Appendix to Sekhri 2014).[16]

B. *Spatial Variation in Expansion*

High-yield crop varieties were not uniformly successful across India. As discussed in more detail in Bharadwaj et al. (2020), the adoption of high-yielding varieties depended on various climatic and environmental conditions. We exploit the sensitivity of high-yield varieties to water availability—specifically, groundwater access. Agriculture in some areas still relied on rain, and investments in canal irrigation had been underway for centuries (Asher et al. 2022). However, since high-yield varieties were particularly sensitive to the timing of irrigation, underground aquifers enabled farmers to better control the timing and amount of irrigation, explaining the faster expansion of the Green Revolution in districts with better access to groundwater (Sengupta and Ghosh 1968). In the years after 1966, acreage devoted to high-yield varieties increased substantially in regions with the greatest endowments of groundwater (Ramasamy, Paramasivam, and Kandaswamy 1994; Sekhri 2014).

The thickness of aquifers was determined prehistorically. Jain, Agarwal, and Singh (2007) document the age of various aquifers, their rock composition, and their spatial coverage in India. The earliest Indian aquifer formations, from the pre-Cambrian era, date back to 3,500 million years ago. The late pre-Cambrian formations are 600 to 1,400 million years old. A tertiary system of aquifers in Northern India was formed from Himalayan upheavals. The youngest aquifers at the foothills of the Himalayas are of the Pleistocene age, 2 million to 11,000 years ago. Most of Central India is under the Deccan Traps, from the late Cretaceous to early Eocene eras. The thickness of these rock formations ranges from 10–50 meters in the eastern part of the country to 2,000 meters in the west (Singhal 1997). These different types of formations are responsible for different hydrological characteristics of the various aquifers. Much of the Deccan Plateau has sporadic aquifers, whereas most of the Indo-Gangetic Plain has thicker aquifers. Our identification strategy exploits

[15] Both rice and wheat varieties were introduced at this time, as discussed in Gollin, Hansen, and Wingender (2021) and Bharadwaj et al. (2020). Munshi (2004) notes that the adoption of high-yield varieties for rice was slower than for wheat, but we confirm in our data that adoption still began around 1966.

[16] The introduction and adoption of high-yield crop varieties has continued since 1966 (Gupta, Ponticelli, and Tesei 2021), but high-yield varieties of staples such as rice and wheat have been available in India since 1966.

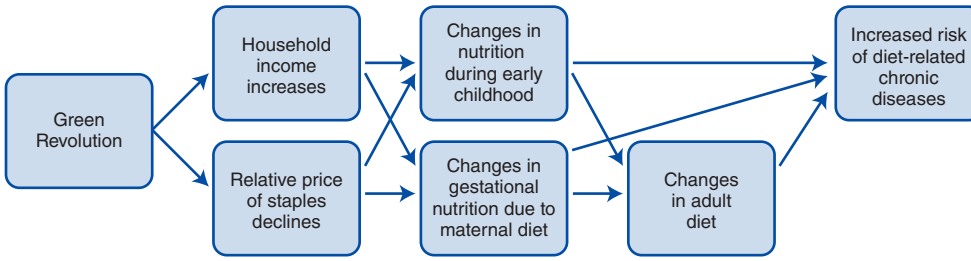

Figure 1. Theory of Change

Note: This figure depicts possible diet-related channels for how exposure to high-yield staple crops while in utero and during early childhood could affect long-run health outcomes.

the exogeneity of the historical, hydrogeological development of aquifer thickness. We address with other possible endogeneity in Section IV below.

II. Expected Effects of High-Yield Staples

Figure 1 depicts the dietary mechanism through which in utero and early childhood exposure to the Green Revolution affects chronic disease. The most direct consequence of the Green Revolution—i.e., the greater crop yields—increased income for farmers. This income shock could affect long-term health through various mechanisms. We focus on the effect of in utero and early childhood exposure to nutritional changes via the contemporaneous dietary habits of pregnant women and very young children.[17]

The impact of this income shock on contemporaneous dietary habits is ambiguous. If staples are normal goods, then an increase in income would increase their consumption. Possibly exacerbating this increase in staple consumption due to the income effect is the decline in the relative price of staples due to increased supply (Pingali 2012).[18] We assume that markets at this time were not fully integrated; changes in local production would have affected local prices more than prices in other districts, particularly in rural areas. Given this, both income and price effects could lead to an increase in the consumption of staples relative to more nutritious foods. Greater income could also increase the consumption of sugar. Alternatively, a large persistent increase in income could lead to dietary diversification and an increase in the relative consumption of protein or vegetables. If staples are an inferior or Giffen good, as found by Jensen and Miller (2008) in China, then staple consumption could fall relative to other foods. The fall in the production of pulses

[17] We acknowledge other possible channels, such as greater investments in infants or less maternal stress (Evans and Garthwaite 2014; Baranov et al. 2020), but since these mechanisms should improve long-term outcomes, we can think of our results as net of these other effects. In Section VI, we consider other changes in the in utero or early childhood environment.

[18] Changes in farming practices may also affect the diversity in available food and relative prices. For example, Cagauan and Arce (1992) note that farmers in the Philippines stopped harvesting wild leafy vegetables and fish from rice paddies due to the increased need for chemical pesticides.

after the Green Revolution (documented in online Appendix C) is inconsistent with this possibility, but the theoretical effect on maternal diet is ambiguous.

These changes in contemporaneous dietary composition can, in turn, affect chronic disease in multiple ways. According to the Barker hypothesis, nutritional conditions in utero or during early childhood can impact long-term health epigenetically (Barker 1990; Harding 2001). One theory is that poor nutrition can lead to the body developing a "thrifty phenotype," which allows it to make efficient use of the calories available. This efficiency can be detrimental, leading to metabolic syndrome, if calories are more abundant. This hypothesis could explain an increase in chronic disease for cohorts born *after* the Green Revolution only in the unlikely event that the rate of growth of calories was such that the mismatch between nutritional needs set in utero and available food manifested itself more for cohorts born after the Green Revolution.[19]

However, another way that in utero nutrition can lead to an increase in chronic disease is through the composition of maternal diet; a low-protein maternal diet has been found to lead to high blood pressure and glucose intolerance in rats and mice (Dahri et al. 1991; Samuelsson et al. 2008; Nivoit et al. 2009; Stocker, Arch, and Cawthorne 2005). These animal studies document the risks of high-fat, high-calorie, and low-protein diets as well as the risk of maternal obesity. King et al. (2014) also document that a postweaning diet high in fat and sugar can affect later glucose tolerance. The literature on humans is supportive of the maternal dietary effect on offspring glucose intolerance but is primarily observational, and most papers focus on maternal obesity (Reynolds et al. 2013; Godfrey et al. 2017; Fleming et al. 2018), while a few study micronutrients (Rao et al. 2001; Yajnik et al. 2008). Waterland et al. (2010) and Dominguez-Salas et al. (2013) show that seasonal dietary changes in Gambia can have epigenetic effects related to diabetes.

It is also possible that in utero nutrition can affect adult diet via different food preferences later in life (Portella et al. 2012). Given the connection between adult diet and diseases such as heart disease and diabetes (Ludwig 2002; Hu and Willett 2002; Hu 2011; Misra et al. 2011; Mohan et al. 2009), this could explain the increase in the chronic disease burden. Specifically, higher carbohydrate intake and lower protein intake in adulthood can increase the likelihood of metabolic syndrome.

In sum, the long-run impact of the Green Revolution on adult health is ambiguous. An improved nutrition environment during gestation and early childhood would be expected to improve adult health, unless changes in dietary composition (either maternal, affecting in utero nutrition, or in early childhood) counter those benefits. Our findings of increased incidence of metabolic syndrome is consistent with an increase in consumption of carbohydrates relative to protein after the Green Revolution.

[19] Luke et al. (2021) adapt the Barker hypothesis to explain the dual burden of malnutrition in many developing countries during periods of rapid economic development, where high rates of malnutrition and increasing rates of metabolic disease coexist.

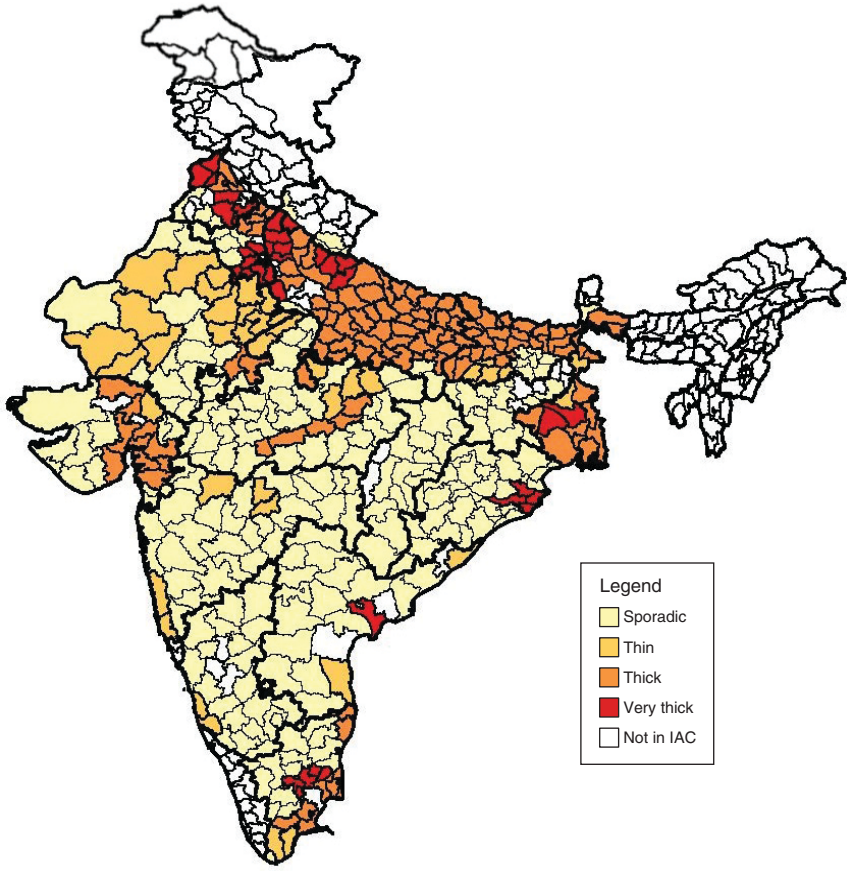

Figure 2. Map of Indian Districts in IAC Shaded by Aquifer Thickness

Notes: This map displays Indian districts in the IAC shaded by aquifer thickness. Unshaded districts were not included in the IAC. Thin borders indicate district boundaries, while thick borders indicate state boundaries. Shapefiles are originally sourced from IPUMS (Minnesota Population Center 2020) with manual edits to the boundaries of Jammu and Kashmir.

III. Data

A. *Aquifer Depth, Climate, and Agricultural Data*

The historical data on adoption of high-yield varieties is from the India Agriculture and Climate Dataset (IAC), assembled by the World Bank (Matlon 1999). The data span 1956 to 1987 and cover 271 of the then-334 districts, focused on 13 major Indian states (Andhra Pradesh, Bihar, Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Orissa, Punjab, Rajasthan, Tamil Nadu, Uttar Pradesh, and West Bengal). The data provide information on acreage devoted to high-yield and conventional varieties of various crops, use of inputs such as fertilizer, edaphic variables including soil quality, and geographical variables such as altitude and distance from the sea.

The data also report the depth of underground aquifers for each district. It classifies Indian districts into four categories—very thick (aquifers greater than 150

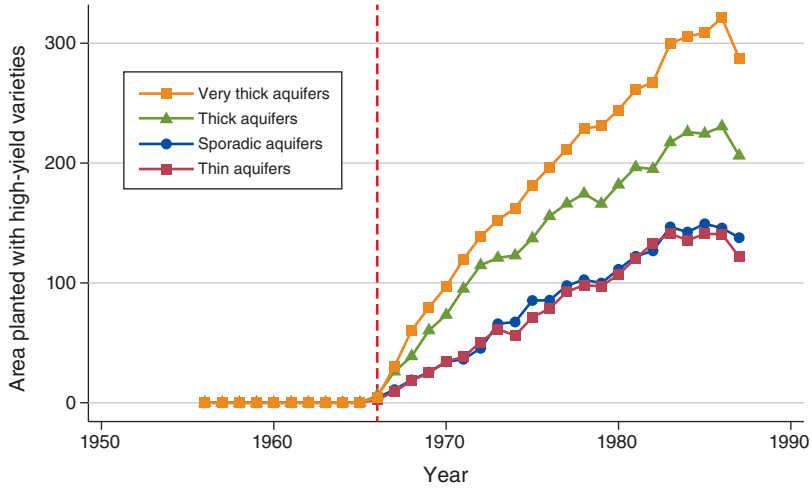

FIGURE 3. ADOPTION OF HIGH-YIELD VARIETIES

Note: This figure graphs the mean area planted with high-yield crop varieties in thousands of hectares, separately for districts with very thick aquifers (yellow), thick aquifers (green), sporadic aquifers (blue), and thin aquifers (red).

meters), thick (aquifers between 100 and 150 meters), thin (aquifers up to 100 meters), and sporadic (very little groundwater availability). Figure 2 presents a map of Indian districts in the IAC data, shaded by aquifer thickness. Approximately 8 percent of the districts have very thick aquifers, 27 percent have thick aquifers, 11 percent have thin aquifers, and 54 percent have sporadic aquifers. Figure 3 plots the average area planted with high-yield varieties over time, separately by groundwater availability. Districts with thick and very thick aquifers saw much faster growth in high-yield adoption than districts with thin or sporadic aquifers.[20] Thus, after confirming this analytically below, we compare water-abundant districts (thick and very thick aquifers) to water-scarce districts (thin or sporadic aquifers). More details describing these data and our process for verifying the data using various government publications are available in online Appendix A (National Atlas & Thematic Mapping Organisation 1984; Ministry of Water Resources 1984, 1997). We also collected average annual rainfall and temperature data from the University of Delaware (Willmott and Matsuura 2001).

B. *Household Health Data*

We use two different surveys to analyze health outcomes, since each has its advantages and limitations. The first data source is the IHDS conducted in 2004–2005 (Desai, Vanneman, and National Council of Applied Economic Research 2005). The IHDS is a nationally representative survey with a sample of 41,554 households located in 33 states and union territories. The IHDS sampled from 382 districts of

[20] Figure 3 includes all high-yield varieties, of which rice and wheat were the most common, but we confirm that adoption grew faster for both rice and wheat in groundwater-rich districts (online Appendix Figure B1).

the then-593 districts in India. Online Appendix Figure B2 presents a map of the districts in the IHDS, shaded by aquifer thickness. The survey collected information on income, consumption, employment, and health. We use responses to whether an individual was ever diagnosed by a doctor with heart disease, diabetes, high blood pressure, cancer, asthma, tuberculosis, cataract, or polio. Given the low frequency of these diseases, and due to concerns about multiple hypothesis testing, we generate a metabolic syndrome index, following Hoynes, Schanzenbach, and Almond (2016), by averaging standardized z-scores of diabetes, heart disease, and high blood pressure for men and adding obesity for women since the IHDS measured weight only for women in 2005.[21] We begin with the IHDS as our primary source of data because it enables us to carefully examine pre-trends using cohorts born prior to 1966.

The second data source is the NFHS-4, another nationally representative household survey, conducted in 2015–2016 (International Institute for Population Studies Mumbai 2017b). The NFHS offers a few advantages over the IHDS. First, and most significantly, it collects blood glucose and blood pressure levels for adults rather than relying on self-reported indicators of diagnoses that may suffer from reporting errors and recall bias. Second, it covers a higher fraction of Indian districts (574 of the then-640 districts). The NFHS has a substantially larger sample of households (628,892), but the men's module was administered to a subset of these households. The number of men in our NFHS sample is twice that of the IHDS. The collection of the data in 2015 is an advantage since chronic disease rates rise with age, but the NFHS only surveyed women aged 15–49 and men aged 15–54.[22] Thus, the main disadvantage of this data source is that we have fewer cohorts of men born prior to the Green Revolution relative to the IHDS (only 10 percent of the sample) and no data on any women born prior to the Green Revolution. More details about these data and construction of outcome variables is available in online Appendix A.

One downside to using survey data is age heaping. As is common in household surveys, respondents in the IHDS report age, from which we infer birth year. Since our identification strategy relies on knowing when individuals were born relative to the start of the Green Revolution, errors in self-reported age (such as rounding to the nearest multiple of five) can be a problem.[23] As described in the next section, we focus on specifications that are less sensitive to this age heaping, but we also confirm that age heaping is uncorrelated with exposure to the Green Revolution and take advantage of the two waves of the IHDS to exclude observations with suspect age reports (Desai, Vanneman, and National Council of Applied Economic Research 2012; please see Section A2 of the data online Appendix for more details).

Tables 1 and 2 report summary statistics for relevant variables for individuals born prior to 1966. These individuals would not be affected by in utero impacts of the Green Revolution but may still be affected by changes in dietary composition later in childhood (the oldest of these cohorts would have been 15 years old when

[21] Obesity is defined as a BMI $\geq$ 30 $\mathrm{kg/m^2}$.

[22] Online Appendix Figure B3 provides a visual of the timeline of events and the age of the first treated cohort in each survey (38–39 years in the IHDS and 49–50 in the NFHS). The IHDS also collected a second wave of data in 2011–2012, but the NFHS offers a larger sample and superior measures of adult health.

[23] Age heaping is not as big a problem in the NFHS, because surveyors asked for both birth date and age and corrected any discrepancies.

Table 1—IHDS Summary Statistics, Restricted to Men Born before 1966

	Observations (1)	All (2)	Sporadic or thin (3)	Thick or very thick (4)	Normalized difference (5)	p-value of difference (6)
Panel A. Individual-level variables						
Age	13,025	45.998	45.983	46.039	0.013	0.549
Literate	13,003	0.745	0.746	0.741	−0.013	0.794
Married	13,024	0.986	0.988	0.981	−0.057	0.037
Metabolic syndrome index	13,024	−0.047	−0.048	−0.045	0.004	0.902
Diabetes	13,024	0.021	0.020	0.025	0.034	0.301
Heart disease	13,024	0.010	0.010	0.011	0.005	0.837
High blood pressure	13,024	0.026	0.027	0.022	−0.031	0.324
Cancer	13,024	0.002	0.002	0.002	0.000	0.983
Asthma	13,024	0.008	0.007	0.009	0.016	0.407
Tuberculosis	13,024	0.006	0.005	0.008	0.040	0.080
Cataract	13,024	0.006	0.005	0.007	0.031	0.194
Polio	13,024	0.001	0.001	0.001	0.018	0.469
Panel B. Household-level variables						
Migrated in	13,010	0.301	0.317	0.269	−0.104	0.121
Urban residence	13,025	0.389	0.377	0.446	0.141	0.160
Eats more rice than wheat	13,025	0.630	0.647	0.559	−0.182	0.108

Notes: This table presents summary statistics from the IHDS. Each row provides the number of observations, the overall mean, the mean for sporadic or thin aquifer districts (the control group), and the mean for thick or very thick aquifer districts (the treated group). Panel A provides individual-level variables, while panel B provides means for household-level variables, restricting the sample to men born before 1966. Column 5 presents the normalized difference between the two types of districts (the difference divided by the standard deviation across the districts), and column 6 presents the p-value of the difference using standard errors clustered by district.

the Green Revolution began). In the IHDS (Table 1), men in groundwater-rich districts are slightly less likely to be married and slightly more likely to be diagnosed with tuberculosis, but these differences are small in magnitude. Households in these districts are less likely to be rice eating (p-value $= 0.108$). In the NFHS (Table 2), men in groundwater-rich districts are slightly less likely to be diagnosed with diabetes (although the difference in glucose levels is not significant) and less likely to have elevated blood pressure. These differences motivate our choice of models and falsification tests.

IV. Empirical Model

Our identification strategy relies on comparing cohorts born after 1966 to those born before 1966 across areas of different historically determined groundwater endowments.[24]

We begin with a simple DID model to study the adoption of high-yield varieties in water-abundant districts:

$$Y_{dt} = \theta(post_t \times W_d) + \beta' X_{dt} + \tau_t + \tau_d + \varepsilon_{dt}, \tag{1}$$

[24] This strategy of comparing areas with different natural resource endowments has been used to examine reductions in poverty in India (Sekhri 2014), to estimate returns to oil production in the United States (Michaels 2010), and to study adaptation of agriculture to climate shocks (Hornbeck and Keskin 2014).

Table 2—NFHS Summary Statistics, Restricted to Men Born before 1966

	Observations (1)	All (2)	Sporadic or thin (3)	Thick or very thick (4)	Normalized difference (5)	p-value of difference (6)
Diabetic according to glucose	7,798	0.262	0.267	0.249	−0.041	0.164
Diabetes diagnosed	8,037	0.063	0.066	0.053	−0.052	0.099
Heart disease diagnosed	8,037	0.030	0.032	0.025	−0.040	0.137
High bl. press. according to readings	7,832	0.631	0.643	0.599	−0.091	0.005
High blood pressure diagnosed	7,831	0.140	0.144	0.131	−0.036	0.291

Notes: This table presents summary statistics from the NFHS. Each row provides the number of observations, the overall mean, the mean for sporadic or thin aquifer districts (the control group), and the mean for thick or very thick aquifer districts (the treated group). The sample includes men born before 1966. Column 5 presents the normalized difference between the two types of districts (the difference divided by the standard deviation across the districts), and column 6 presents the p-value of the difference using standard errors clustered by district.

where Y_{dt} is the area in district d in year t planted with high-yield crop varieties, $post_t$ is an indicator for 1966 or later, and W_d is either a vector or a dummy variable indicating aquifer thickness. X_{dt} are time-varying district characteristics such as average annual rainfall and temperature. τ_t and τ_d indicate year and district fixed effects to control for year-specific shocks and time-invariant district characteristics, respectively. Robust standard errors are clustered at the district level for all regressions since high-yielding varieties were distributed at the district level (Frankel 1971). High-yield varieties were not available in India prior to 1966—the area planted is 0 for all districts, as seen in Figure 3; thus, there are no differential trends prior to the Green Revolution.

We estimate a similar model using individual-level data on health outcomes. Here, however, we might worry about preexisting trends that differ across districts with different groundwater endowments, invalidating our identifying assumption. We test for differential pre-trends by comparing cohorts born after 1956 and 1961 with those born after 1951. We group together multiple cohorts because of the age heaping in the IHDS. Specifically, we estimate

$$(2) \qquad Y_{idt} = \theta(post_t \times W_d) + \theta_{56}(post56_t \times W_d) + \theta_{61}(post61_t \times W_d) + \tau_t + \tau_d + \varepsilon_{idt},$$

where Y_{idt} is a health outcome for individual i in district d born in year t, $post_t$ is an indicator for being born in 1966 or later, and W_d is an indicator for aquifer thickness. The indicators $post56_t$ and $post61_t$ refer to being born in or after 1956 and 1961, respectively; since we limit the sample to those born between 1951 and 1981, the omitted cohorts are those born between 1951 and 1955. We use the sample weights available in the data. In equation (3), we include state × birth-year fixed effects (τ_{st}) and district-specific trends in birth year ($t\tau_d$) instead of the falsification tests:

$$(3) \qquad Y_{idt} = \theta(post_t \times W_d) + \tau_t + \tau_d + \tau_{st} + t\tau_d + \varepsilon_{idt}.$$

We also use an event study model to flexibly estimate time-varying effects of the Green Revolution, conditional on birth-year, district, and state × birth-year fixed effects and on district-specific trends. The event study approach allows for the effect

Table 3—Adoption of High-Yield Varieties

Variables	HYV area (1)	HYV area (2)
Post-1966		
× thin aquifers	−4.84	
	(12.02)	
× thick aquifers	55.44	
	(13.58)	
× very thick aquifers	104.41	
	(20.83)	
× thick or very thick aquifers		67.59
		(11.85)
Observations	8,640	8,672
R^2	0.73	0.72

Notes: Each column presents the results from estimating equation (1) using the area planted with high-yield crop varieties (HYV) as the dependent variable. All regressions include district and year fixed effects as well as controls for average annual rainfall and average annual temperature. Standard errors clustered at the district level are in parentheses.

to grow over time, diminish over time, or even change nonmonotonically. It also allows us to support our identifying assumption by flexibly estimating differential trends in health outcomes prior to the Green Revolution. In order to address the age heaping, we group birth cohorts into three-year bins and estimate the evolution of differences across districts, relative to the omitted group born between 1963 and 1965. We define four groups born before 1966 (1951–1953, 1954–1956, 1957–1959, and 1960–1962) and five groups born after 1966 (1966–1968, 1969–1971, 1972–1974, 1975–1977, 1978–1980). We estimate

$$(4)\quad Y_{idt} = \sum_{l=1}^{4} \theta_l^{pre}(\tau_l \times W_d) + \sum_{l=6}^{10} \theta_l^{post}(\tau_l \times W_d) + \tau_t + \tau_d + \tau_{st} + t\tau_d + \varepsilon_{it},$$

where τ_l are dummy variables indicating age group. θ_l^{pre} and θ_l^{post} are streams of coefficients estimated for the "pre-" cohort groups and the "post-" cohort groups. Estimates of θ_l^{post} tell us how the impact of the Green Revolution changes over time, and estimates of θ_l^{pre} close to 0 support our identifying assumption.

V. Results

A. *Adoption of High-Yield Varieties*

We estimate equation (1) for the area planted with high-yield varieties (in thousands of hectares) and present the results in Table 3. Column 1 compares districts in all four aquifer thickness categories (the omitted category is districts with sporadic aquifers). As previewed in Figure 3, the difference in adoption between districts with thin and sporadic aquifers is small and not statistically significant, while the districts with thick and very thick aquifers saw substantially greater adoption. Column 2 groups together the thick and very thick aquifer districts, comparing them with the sporadic and thin aquifer districts. Water-abundant districts planted 68,000

Table 4—Difference-in-Differences Estimates for Long-Term Health

Variables	Metabolic syndrome index (1)	Diabetes (2)	Heart disease (3)	High blood pressure (4)
Difference-in-differences				
Born $\geq$ 1966 $\times$ groundwater rich	0.056	0.011	0.0071	0.0084
	(0.021)	(0.0047)	(0.0041)	(0.0048)
Pre-trends (*rel. to 1951–55 cohorts*)				
Born $\geq$ 1956 $\times$ groundwater rich	−0.033	−0.021	−0.00056	0.0063
	(0.037)	(0.013)	(0.0080)	(0.010)
Born $\geq$ 1961 $\times$ groundwater rich	−0.048	−0.0069	−0.0098	−0.0046
	(0.027)	(0.0064)	(0.0044)	(0.0068)
Observations	35,160	35,160	35,160	35,160
R^2	0.057	0.042	0.031	0.038

Notes: Each column presents the results from estimating equation (2) using a health outcome from the IHDS as the dependent variable. The dependent variables in columns 2–4 indicate whether the individual was diagnosed with these conditions. The dependent variable in column 1 is an index of metabolic health, averaging standardized z-scores of the three conditions. The sample is restricted to men born between 1951 and 1981. Individuals with substantially discrepant reported ages across the two waves of the IHDS are excluded (see text for more details). All regressions include district and birth-year fixed effects and use sampling weights. Standard errors clustered at the district level are in parentheses.

more hectares of high-yield crops after 1966 than did water-scarce districts. From here on, we group together districts with thick or very thick aquifers, referring to them as groundwater-rich districts. We confirm these results using an event study model, flexibly estimating the differential growth of high-yield adoption (online Appendix Figure B4). The area planted with high-yield varieties grew significantly faster in water-abundant districts after 1966. Separate estimates for rice and wheat confirm significant differential growth for both staples after 1966.

B. *Long-Term Health*

Men in the IHDS, 2004–2005.—Table 4 estimates equation (2) to study the impact of the Green Revolution on diet-related diseases for men. Column 1 uses the metabolic syndrome index as the dependent variable, while indicators for the three components—diabetes, heart disease, and high blood pressure—are presented in columns 2–4. The first row indicates the DID estimate between groundwater-rich and groundwater-scarce districts for cohorts born after 1966. We see evidence of increases in the index and diabetes prevalence, and marginally significant increases in heart disease and high blood pressure prevalence. While these regressions control for fixed differences across districts and across years, interpreting these estimates as the impact of the Green Revolution requires assuming parallel trends between groundwater-rich and groundwater-poor districts. The second and third rows of Table 4 test this assumption with similar DID estimates entirely between cohorts born prior to the Green Revolution, comparing those born after 1956 or 1961 with those born between 1951 and 1955. This exercise serves as a falsification test since none of these cohorts were exposed in utero

Table 5—DID Estimates for Long-Term Health with Additional District and State Controls

Variables	Metabolic syndrome index (1)	Diabetes (2)	Heart disease (3)	High blood pressure (4)
Born $\geq$ 1966 $\times$ groundwater rich	0.13	0.039	0.011	0.013
	(0.062)	(0.016)	(0.012)	(0.014)
Observations	35,085	35,085	35,085	35,085
R^2	0.122	0.118	0.097	0.096

Notes: Each cell presents the results from estimating equation (3) using a health outcome from the IHDS as the dependent variable. The dependent variables in columns 2–4 indicate whether the individual was diagnosed with these conditions. The dependent variable in column 1 is an index of metabolic health, averaging standardized z-scores of the three conditions. The sample is restricted to men born between 1951 and 1981. Individuals with substantially discrepant reported ages across the two waves of the IHDS are excluded (see text for more details). All regressions include district, birth-year, and state $\times$ birth-year fixed effects. They also include district-specific trends and use sampling weights. Standard errors clustered at the district level are in parentheses.

to the Green Revolution. Interestingly, while we see a few significant differences among cohorts born prior to the Green Revolution, almost all the coefficients—and certainly the significant ones—are negative: among those born prior to the Green Revolution, younger individuals in groundwater-rich districts are less likely to be diagnosed with these diseases. If anything, differential pre-trends appear to be going in the opposite direction.

To address these differential pre-trends, Table 5 estimates equation (3) with district-specific trends and state $\times$ birth-year fixed effects. We see corroborating evidence of an increase in the metabolic syndrome index and diabetes. The estimates for heart disease and high blood pressure are of larger magnitude but not statistically significant. We conclude that men born after the Green Revolution experience higher risk of metabolic syndrome in water-abundant districts relative to men in water-scarce districts. The coefficients for the individual components are easier to interpret: prevalence of diabetes is about 4 percentage points higher for these men. We do not find evidence of an increase in heart disease and high blood pressure, but this could be due to a lack of statistical power, given the rarity of these diseases at this young age. Online Appendix Table B1 confirms that these effects are robust to (i) clustering by state instead of district, (ii) not using sample weights, (iii) omitting 40-year-old men due to the age heaping and the greater uncertainty around whether those who report being 40 were born before or after the Green Revolution, (iv) adding household fixed effects, (v) controlling for an array of district characteristics interacted with birth-year indicators, and (vi) controlling for various individual characteristics interacted with groundwater availability.[25] Online Appendix Table B2 confirms that our estimate of the impact of the Green Revolution on the

[25] The district characteristics included are, for those born prior to 1966, district averages of literacy, migrant status, whether households eat more rice or wheat, educational attainment (any, completed primary, completed middle school, completed high school, any tertiary), and religion (Muslim, other, the omitted group being Hindu). The individual characteristics included are literate, married, migrant, education categories, religion categories, and caste categories.

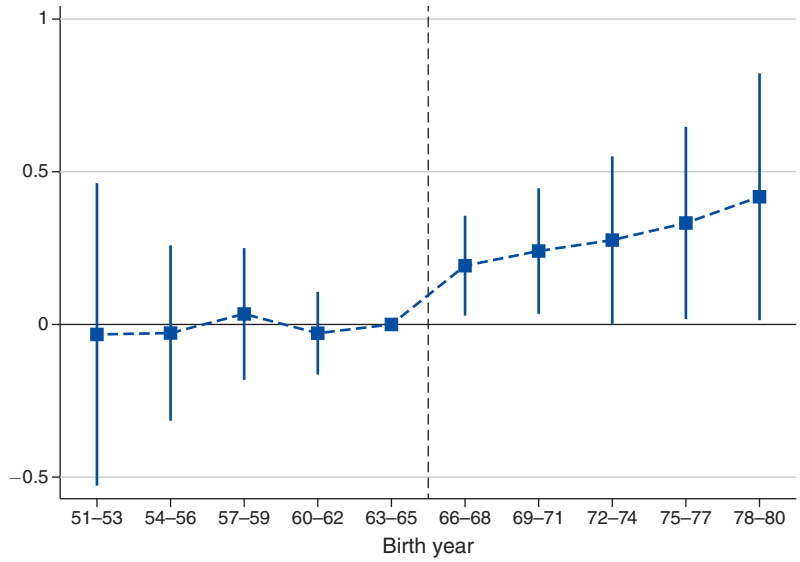

Figure 4: Event Study Estimates for Impact on Metabolic Syndrome Index

Notes: This figure plots the coefficients from estimating equation (4) using the metabolic syndrome index as the dependent variable. Those born between 1963 and 1965 are the omitted group. The regression includes district, birth-year, and state × birth-year fixed effects. It also includes district-specific trends and uses sampling weights. Vertical bars indicate 95 percent confidence intervals.

metabolic syndrome index is robust to using principal component analysis or factor analysis to generate the index.

Figure 4 presents estimates from the event study model, equation (4), supporting our conclusion of an increase in the metabolic syndrome index for men; Figure 5 provides a similar figure for diabetes.[26] As described above, we group cohorts into three-year bins. The bars indicate 95 percent confidence intervals. The event studies show a significant increase in diabetes for men born immediately after the Green Revolution, with no evidence of differential trends for those born before, even though these older cohorts would also have been exposed to the Green Revolution.[27] In other words, for those born before the Green Revolution, diabetes risk is unrelated to age of first exposure, conditional on district trends and state × birth-year fixed effects. This is consistent with our strategy identifying the effect of exposure to high-yield varieties in utero or during early childhood.

Threats to Identification.—Our identification strategy of exploiting temporal variation in birth and geographic variation in groundwater abundance, and controlling for district fixed effects, district-specific trends, and state × birth-year fixed effects addresses many possible sources of bias. Indian economic liberalization in the 1990s

[26] Event study figures for heart disease and high blood pressure (online Appendix Figures B5 and B6) are consistent with the lack of significant results on these components in Table 5.

[27] The pre-1966 coefficients are not jointly significant in either Figure 4 or Figure 5 (p-values 0.92 and 0.21, respectively).

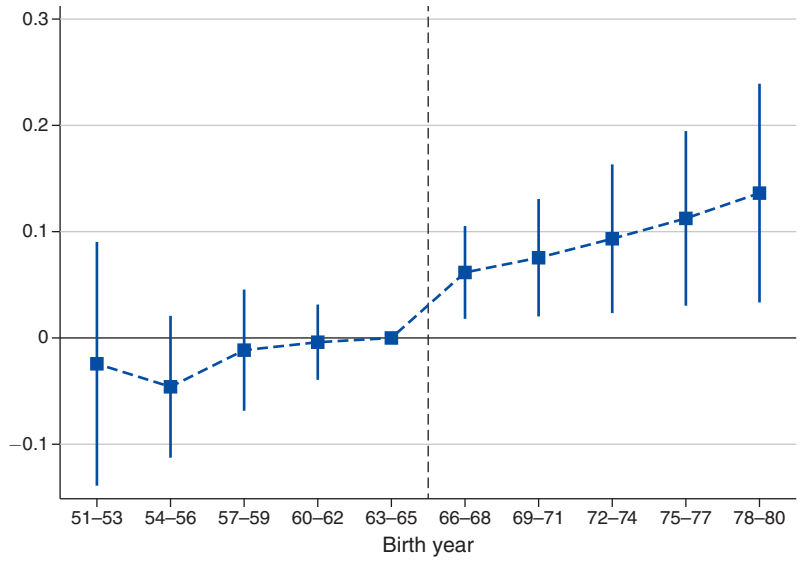

FIGURE 5. EVENT STUDY ESTIMATES FOR IMPACT ON DIABETES

Notes: This figure plots the coefficients from estimating equation (4) using diabetes as the dependent variable. Those born between 1963 and 1965 are the omitted group. The regression includes district, birth-year, and state × birth-year fixed effects. It also includes district-specific trends and uses sampling weights. Vertical bars indicate 95 percent confidence intervals.

and recent trends in dietary habits (more sugary, processed foods), lifestyle (activity levels, sedentary jobs), and health care access should be addressed with these fixed effects since even if these trends are correlated with groundwater availability, they should affect people born before 1966 as well as those born after 1966. Similarly, our strategy should address the possibility of differential migration even though we use district of current residence to determine exposure (we do not know district of birth). Even if migration patterns are related to both aquifer thickness and risk of metabolic syndrome, they would only drive diabetes for those born after 1966 more than those born before if migration was related to exposure during gestation or early childhood, which seems even less likely. Migration was not very common in India during this time period, but we confirm that migration patterns are unrelated to exposure to high-yield varieties at young ages and that this does not differ by health status (online Appendix Table B3).[28] Remaining threats to identification are difficult to envision since they would have to drive differences in diabetes across districts with varying groundwater access for those born immediately after 1966 but not those born before, even after controlling for state × birth-year fixed effects and district-specific trends.

[28] We use the IHDS question "How many years ago did your family first come to this village/town/city?" to determine which households have migrated to this location, but we do not have data on the district of origin for migrants. We also use tracking data from the second wave of the IHDS to determine whether individuals surveyed in 2005 had moved out prior to 2012. Even if this out-migration was related to district-of-origin aquifer thickness, it would not bias our 2005 estimates, but the lack of a relationship provides additional reassurance that out-migration prior to 2005 is not driving our results.

Table 6—DID Estimates for Long-Term Health Using NFHS Data

Variables	Diabetic according to glucose readings (1)	Diabetes diagnosed (2)	Heart disease diagnosed (3)	High BP according to readings (4)	High BP diagnosed (5)
Panel A. Difference-in-differences					
Born $\geq$ 1966 $\times$ groundwater rich	0.045	0.021	−0.0064	0.00092	−0.0016
	(0.019)	(0.011)	(0.0070)	(0.022)	(0.013)
Pre-trends (rel. to 1960–1962 cohorts)					
Born $\geq$ 1963 $\times$ groundwater rich	0.018	−0.00092	0.0076	0.022	−0.017
	(0.029)	(0.020)	(0.016)	(0.031)	(0.024)
Observations	76,664	79,197	79,197	76,951	76,950
R^2	0.046	0.041	0.026	0.069	0.091
Panel B. District and state controls					
Born $\geq$ 1966 $\times$ groundwater rich	0.054	0.0068	−0.011	0.014	0.0014
	(0.031)	(0.013)	(0.0071)	(0.030)	(0.025)
Observations	76,648	79,183	79,183	76,936	76,935
R^2	0.072	0.076	0.052	0.094	0.123

Notes: Each column presents the results from estimating equation (2) in panel A and equation (3) in panel B using a health outcome from the NFHS as the dependent variable. The dependent variables in columns 1 and 4 indicate whether the individual has diabetes or high blood pressure (BP) according to biomarker readings. The dependent variables in columns 2, 3, and 5 indicate whether the individual was diagnosed with one of these conditions. The sample includes men born between 1960 and 1992. All regressions include district and birth-year fixed effects and use sampling weights. Panel B includes district-specific trends and state $\times$ birth-year fixed effects. Standard errors clustered at the district level are in parentheses.

Men in the NFHS, 2015–2016.—We next verify our results using the NFHS from 2015–2016. An important benefit of the NFHS is that the survey collected blood glucose and high blood pressure readings in addition to self-reported data on diagnoses of these diseases. This round of the NFHS was conducted 11 years after the IHDS, when these diseases are likely to be more prevalent among the (now-older) cohorts born around the time of the Green Revolution.[29] However, the NFHS only surveyed men aged 15–54. Thus, less than 10 percent of the sample was born before 1966, and most of those between 1963 and 1965. One consequence is that we are able to test for pre-trends in these data with only a few cohorts. We estimate specifications similar to equation (2), using only one falsification group (those born between 1963 and 1965, where the omitted group is those born between 1960 and 1962).

Panel A of Table 6 presents results for indicators of diabetes, heart disease, and high blood pressure. In column 1, we see that those born after 1966 in water-abundant districts are 4.5 percentage points more likely to have blood sugar levels in the diabetic range than those born before. The point estimate for the falsification test—that is, the DID estimate for those born between 1963 and 1965, as compared to those born before 1963—is smaller and not statistically significant. This is confirmed by a

[29] While the IHDS revisited sample households in 2011–2012, also offering the opportunity to study affected cohorts at a later age, the NFHS dominates due to its collection of biomarkers. Estimates using the second wave of the IHDS are imprecise, possibly due to higher mortality of men with diabetes (online Appendix Table B4).

2 percentage point increase in diabetes diagnoses, also statistically significant (column 2). Mirroring the results from the IHDS, columns 3–5 find no evidence of an effect of the Green Revolution on heart disease (diagnosis) or high blood pressure (readings or diagnosis). Panel B of Table 6 estimates specification (3) adding district-specific linear trends and state × birth-year fixed effects. The impact of the Green Revolution on blood sugar levels is robust to these additional controls—the magnitude of the coefficient is greater—although the effect is only marginally significant. These results lend credence to the results in the IHDS: we see an increase in diabetes prevalence among men born after the Green Revolution in water-abundant districts.

Women in the IHDS, 2004–2005.—Repeating this analysis for women, we find no evidence of differential changes in the risk of metabolic syndrome or diabetes for those born after the Green Revolution (see online Appendix Table B5). Recall that we are only able to use the IHDS to study women because the NFHS only surveyed women aged 15–49, all of whom were born after 1966.

We posit a number of possible explanations for the lack of an effect for women. One likely explanation is biological differences in the age gradients of diabetes prevalence for women versus men (recall that our IHDS sample is 24–54 years old, with those born after the Green Revolution younger than 39 in 2005). If women are at higher risk of metabolic syndrome after menopause (Carr 2003; Kautzky-Willer, Harreiter, and Pacini 2016; Paschou et al. 2019; Polotsky and Polotsky 2010), it may be too early to see the effect for women (the average age of menopause in India is 46 according to Ahuja 2016). To the best of our knowledge, there are no publicly available data sources with information on chronic diseases for older women that contain district identifiers. It is also possible that there are biological differences in how boys and girls respond to similar in utero or early childhood dietary environments. Both Bharadwaj et al. (2020) and von der Goltz et al. (2020) find larger impacts of the Green Revolution on infant mortality for males than for females.

A second explanation is gender disparities in intrahousehold distribution of food (during adulthood) or other child investments during early childhood. Anecdotally, female household members in India often eat after male household members finish their meals (Sedlander et al. 2021) and often eat different food (for example, women are more likely to be vegetarian, as noted in Yadavar 2018). Related, D'Agostino (2017) finds that the Green Revolution in India increased the rural gender wage gap since the productivity gains in farming mostly accrued to men. It is possible that this affected gender differences in child investments. A distinct but related explanation is gender differences in health care utilization and diagnosis. Since the NFHS does not have data on older women, we are unable to test for an impact on blood sugar levels.

Finally, a third explanation is that we have less statistical power for women. Migration of household heads is uncommon in India, but migration of women for marriage is more common. Unfortunately, the IHDS only reports migration status for the household head. Given these various explanations and the lack of data to explore them further, we leave this question to future work.

VI. Mechanisms

A. *Changes in Diet*

As discussed in Section II, household dietary changes around the time of birth could explain the increase in metabolic syndrome for men born after the Green Revolution through changes in nutrition in utero (via a low-protein maternal diet) or during early childhood. Individual-level data on dietary intake is sufficiently scarce that we are unable to document these dietary changes among pregnant women or infants in the 1960s, but we provide a few pieces of suggestive evidence that dietary changes led to the increase in metabolic syndrome. First, we show that the long-term health impacts of the Green Revolution vary for households that differ in their historically determined preferred staple (rice or wheat) in ways that are consistent with these health effects. Second, we use recent household-level data to compare consumption patterns of households with adult men born before and after the Green Revolution across districts. Finally, in the next subsection, we offer evidence ruling out other possible explanations.

Typical Indian diets are based on two staples—rice and wheat—but households differ in which one is the primary staple. As explored further in Atkin (2013, 2016), this dietary habit is stable over time, and while it has regional and cultural roots, the preference survives migration to other regions of India. In fact, Atkin (2013) argues that this preference has consequences for calorie consumption. Rice also has a higher glycemic index (measuring how quickly foods affect blood sugar) and less protein than wheat. Given the possible consequences of a low-protein maternal diet discussed in Section II and the risks of foods with high glycemic index for diabetes,[30] we might expect the impact of exposure to the Green Revolution on diabetes to differ across rice-eating and wheat-eating households. In fact, some medical studies have linked white rice to diabetes, particularly for the South Asian population (Bhavadharini et al. 2020; Golozar et al. 2017), while others have found suggestive—but more mixed—results in other populations (Hu et al. 2012, Seah et al. 2019, Nanri et al. 2010).

Table 7 uses equation (3) to estimate heterogeneous effects of the Green Revolution on diabetes by including interactions of our independent variable of interest (being born after 1966 in groundwater-rich districts) with an indicator for whether the household's primary staple is rice. We infer this household characteristic from expenditure data gathered by the IHDS. Roughly half of the men in our sample live in households that consume more rice, while the other half live in households that consume more wheat.[31] Column 1 in Table 7 demonstrates that the effect on diabetes is larger for households that eat more rice than wheat (the difference is marginally significant, at the 10 percent level).[32] Since rice eating is a regional

[30] A recent theory links diabetes more closely to the glycemic load of dietary intake than to calories, although this model of linkages is not universally accepted (Ludwig and Ebbeling 2018; Hall, Guyenet, and Leibel 2018).

[31] Rice eating is unaffected by exposure to the Green Revolution (online Appendix Table B6).

[32] We acknowledge that we are relying on cross-district variation in this heterogeneity analysis; in online Appendix Table B7, we include district × birth-year fixed effects. The coefficients on rice-eating households are positive but not significant; we do not have enough statistical power to rely solely on within-district variation.

Table 7—Heterogeneity Estimates for Diabetes, by Diet

Variables	(1)	(2)
Born $\geq$ 1966 $\times$ groundwater rich	0.0023	0.0024
	(0.0063)	(0.0070)
$\times$ rice-eating household	0.040	
	(0.021)	
$\times$ rice-eating state		0.047
		(0.026)
Observations	34,900	35,085
R^2	0.146	0.118
Prob $>$ F (joint)	0.093	0.15
Prob $>$ F (main + interaction)	0.032	0.059

Notes: Each column presents the results from estimating equation (3) using an indicator for diabetes from the IHDS as the dependent variable. The sample is restricted to men born between 1951 and 1981. Individuals with substantially discrepant reported ages across the two waves of the IHDS are excluded (see text for more details). All regressions include district, birth-year, and state $\times$ birth-year fixed effects. They also include district-specific trends and use sampling weights. All fixed effects and trends are also interacted with the relevant household characteristic. Standard errors clustered at the district level (column 1) and state level (column 2) are in parentheses.

characteristic, we divide states into rice-eating and wheat-eating states based on how many households eat more rice. Column 2 in Table 7 shows that the impact is marginally significantly larger in rice-eating states. This evidence is consistent with at least two possible dietary channels: (i) low-protein diets in rice-eating lineages increases risk of diabetes, or (ii) low-protein diets among all households leads to greater risk of diabetes only among rice-eating men. We are unable to distinguish between these channels, but both suggest a role played by dietary changes.

As noted in Section II, changes in nutrition in utero or during early childhood can theoretically affect adult diet via different food preferences (Portella et al. 2012). Another piece of suggestive evidence comes from household-level data on food consumption from the National Sample Survey Organization's Household Consumption Expenditure Surveys conducted in 1999–2000 (see online Appendix A3 for more details on the data). We estimate the difference in daily per capita consumption of macronutrients (total calories, protein, and fat) of adult males born after 1966 as compared to adult males born after 1961 and after 1956. One caveat is that the data are at the household-level, so the outcome variable is household consumption divided by the number of "consumption units" (in order to account for the fact that women and children eat less than adult men; see online Appendix A3 for more details). We test whether adult males born after the Green Revolution live in households that consume more calories per capita than adult males born before it and whether this difference is larger in groundwater-rich districts. We estimate a model very similar to equation (2) except that we include state $\times$ birth-decade fixed effects. We find that men born after 1966 in groundwater-rich districts live in households that consume more total calories, of which 10.8 come from protein and 46 from carbohydrates (including staples and sugar, online Appendix Table B8). These are not age effects since we include birth-year fixed effects; they are also not

age × district effects since those born after 1956 and those born after 1961 live in households that consume *fewer* calories than those born even earlier (although these differences are not statistically significant). They are unlikely to be simple income effects unless households with adult men born after 1966 are differentially richer in groundwater-rich districts relative to households with slightly older men.

Given current nutritional guidelines linking carbohydrates to diabetes, these results are illuminating. The magnitudes are not trivial—a household with an adult male born after 1966 consumes 77 more calories per person per day if in a district with thicker aquifers. Putting this number in context, Cutler, Glaeser, and Shapiro (2003) argue that an increase of 100–150 calories per day can explain the increase in median weight in the United States.

We also look into how the Green Revolution could have affected diet by examining the impact of the introduction of high-yield varieties on the production of various crops. Given that markets were not fully integrated, local production of different crops likely affected diet. Online Appendix C expands upon this discussion, documenting that production of staples increased and that production of pulses fell in absolute terms in groundwater-rich districts after 1966, not just relative to staples. Pulses are a primary source of protein in the South Asian diet. Production of sugar may also have been affected by the Green Revolution and resulting increases in sugar consumption could also contribute to a low-protein maternal diet. Sugar is a water-intensive crop and, prior to the Green Revolution, groundwater-rich districts produced more sugar. However, in contrast to the rise in staples output and the decline in pulses production, there does not seem to be a sustained or significant differential increase in sugar production after 1966 in groundwater-rich districts (online Appendix Figure C3, panel B and Figure C6).[33] While we cannot rule out that the low-protein diet for pregnant women and young children was driven by increases in sugar consumption, these trends in agricultural output suggest that our diabetes results are driven more by dietary substitution away from pulses toward staples.

B. *Alternative Mechanisms*

Next, we consider two alternative explanations for our findings. First, the Green Revolution might have affected health through in utero or early childhood exposure to other environmental factors (e.g., pollution, fertilizer, water quality). Second, differential survival until 2005 could explain the increase in metabolic syndrome if those who would develop these diseases had already died in water-scarce districts. A related explanation is that the reduction in infant mortality due to the Green Revolution might have affected the composition of surviving children whereby the marginal child who survived in the groundwater-rich districts was more prone to metabolic syndrome than in other districts. In this subsection, we provide evidence arguing against these alternative mechanisms.

[33] If anything, relative sugar production falls in groundwater-rich districts immediately after 1966. We do not have prices or consumption data, but given the production results, we do not believe the price of sugar changed over time differentially.

Other In Utero Exposures.—While our strategy identifies the effect of in utero and early childhood exposure, the Green Revolution could also have affected long-term health in ways unrelated to dietary changes. For example, changes in farming practices resulting in greater exposure to chemical pesticides or fertilizers could also have adverse health impacts.

We might expect such exposure to be greater for men born in rural areas, suggesting that we examine heterogeneous effects by type of location. Unfortunately, dietary changes may also be greater in rural areas, making such heterogeneous results inconclusive. Nevertheless, in online Appendix Table B9 we examine heterogeneous results with respect to rural residence and a proxy for rural origins based on current location and the type of location from which migrant households had most recently moved (the IHDS asks this of household heads who had not resided in the current location since birth). While the increase in diabetes is not significantly different for current rural residents, we see some evidence of a greater increase in diabetes for households with a rural origin (the difference is almost significant with a p-value of 0.103). However, even within households with rural origins, we find that the increase in diabetes is driven by rice-eating households, supporting a dietary driver for our results (see online Appendix Table B9). We acknowledge, however, that we are not fully able to identify whether the main driver is the lower protein content of rice, the higher glycemic index of rice, or an interaction between rice and environmental exposure.

One possible environmental mechanism is DDT exposure. Recent studies have documented a correlation between DDT exposure and diabetes (La Merrill et al. 2019, Cirillo et al. 2021), and as described in Cutler et al. (2010), DDT was used in a campaign to eradicate malaria in India beginning in 1953. By 1960–1961, however, the campaign covered the entire country, and DDT use was ubiquitous; our fixed effects should rule out the impact of DDT usage. Even if the expansion between 1953 and 1960 was correlated with groundwater availability, we would see any impact on diabetes five to six years earlier than we do. In our pre-trend analysis, we see the opposite: men born right after 1961 are actually less likely to be diagnosed with these chronic diseases than men born earlier, and these pre-trends disappear with state $\times$ birth-year fixed effects.

Another possible environmental mechanism is exposure to chemical fertilizer; fertilizer use rose significantly as a result of the Green Revolution. To the best of our knowledge, fertilizer use has not been linked to diabetes, but we cannot fully rule it out as a mechanism. We offer some suggestive evidence that it does not drive our results. Specifically, we show that our results are robust to controlling for fertilizer use in the year of birth and that the amount of fertilizer use itself does not affect prevalence of diabetes (online Appendix Table B10). An increase in exposure to fertilizer also does not explain why the increase in diabetes prevalence is greater for rice-eating households, another result that is robust to fertilizer controls.

Finally, we test the effect of exposure to the Green Revolution during gestation and early childhood on a host of other non-diet-related health conditions. We find no evidence of differential impacts on cancer, asthma, tuberculosis, cataracts, and polio (Table 8). These results also help rule out changes in air pollution due

Table 8—DID Estimates for Long-Term Health Conditions Unrelated to Diet

Variables	Cancer (1)	Asthma (2)	TB (3)	Cataract (4)	Polio (5)
Born $\geq$ 1966 $\times$ groundwater rich	−0.00090	−0.0014	−0.0099	0.0069	0.0018
	(0.0033)	(0.011)	(0.0073)	(0.0058)	(0.0018)
Observations	35,085	35,085	35,085	35,085	35,085
R^2	0.139	0.064	0.061	0.063	0.050

Notes: Each cell presents the results from estimating equation (3) using a health outcome from the IHDS as the dependent variable. The sample is restricted to men born between 1951 and 1981. Individuals with substantially discrepant reported ages across the two waves of the IHDS are excluded (see text for more details). All regressions include district, birth-year, and state $\times$ birth-year fixed effects. They also include district-specific trends and use sampling weights. Standard errors clustered at the district level are in parentheses.

to environmental factors such as rubble burning or arsenic poisoning as the main mechanism, since these are more closely linked to respiratory diseases and cancer.

Differential Survival.—Lastly, we argue that our results cannot be driven by increased survival to adulthood and composition effects. We begin by demonstrating that despite reductions in infant mortality, survival to adulthood in 2005 seems, if anything, lower in groundwater-rich districts. We use data on population by five-year age group and gender from the 2001 census of India (Census of India 2001), four years before the IHDS. If the results were driven by greater survival in districts with more groundwater, we should see more people in those districts who were born after the Green Revolution. We estimate this first using specification (3) except that birth year is in five-year groups (online Appendix Table B11). The dependent variable is the number of men (in thousands) in a district in a five-year age group. Individuals aged 35 and older were all born prior to the Green Revolution, and cohorts 34 and younger were born after. The only statistically significant coefficients are actually of the opposite sign, but not significant with state $\times$ age-group fixed effects.

However, even if survival to 2005 was no different in the different types of districts, a reduction in infant mortality driven by the Green Revolution (Bharadwaj et al. 2020; von der Goltz et al. 2020) could explain our results if the marginal surviving child in water-abundant districts was more susceptible to metabolic syndrome later in life. Such a change in the composition of health endowments would mean that, while the Green Revolution still caused the increase in chronic disease, the driving mechanism would be the reduction in infant mortality rather than dietary changes. In order to explore this, we conduct a worst-case-scenario bounding exercise.

We begin by calculating the expected relative change in infant mortality in groundwater-rich districts due to the adoption of high-yield varieties using estimates from Bharadwaj et al. (2020) for India. They find that a 20 percentage point increase in the fraction of area devoted to high-yield varieties reduces the probability an infant dies by 0.5 percentage points. We first estimate the change in the share of high-yield varieties due to the Green Revolution in our data (online Appendix Table B12). The predicted reduction in mortality is 0.2 percentage points for groundwater-rich districts. We then trim the population of men born

after 1966 by this percentage, focusing on those with a nonzero metabolic syndrome index or with diabetes. In online Appendix Table B13, we select the observations to drop randomly and present the DID results along with a bootstrapped 95 percent confidence interval. Alternatively, in online Appendix Table B14, we drop the poorest men with a nonzero metabolic syndrome index or diabetes (and therefore do not bootstrap). The results are quantitatively similar to our main results.[34]

In sum, we find evidence supporting dietary changes affecting nutrition during gestation and early childhood, and no evidence supporting other environmental changes or survival as explanations for the increase in metabolic syndrome.

VII. Conclusion

In this paper, we examine the long-term impacts on adult health of the Green Revolution in India. We exploit geographic variation in historically determined aquifer thickness that predicts the adoption of high-yield crop varieties. We find increases in the prevalence of diabetes for men born after the start of the Green Revolution in groundwater-rich districts. These negative long-term impacts are surprising given that most of the literature finds long-term health benefits of positive income and nutrition shocks during early childhood. We provide suggestive evidence of a possible mechanism: changes in contemporaneous dietary habits affected nutritional conditions in utero or during early childhood. A maternal diet low in protein can increase the tendency toward diabetes. These dietary changes could have resulted either from income growth or from the abundance and lower relative prices of rice and wheat. To support our hypothesis of a dietary channel, we show that our results are larger for rice-eating households than for wheat-eating households. Rice affects blood sugar faster (has a higher glycemic index) and contains less protein than wheat. We provide evidence ruling out alternative explanations for our results, such as other environmental exposures while in utero and differential survival of those prone to chronic diseases.

These findings have implications for current agricultural policy and welfare policy. For example, the Public Distribution System, India's food security system for poor households, inadvertently promotes a high-calorie, low-protein diet by prioritizing the distribution of wheat, rice, sugar, and kerosene; only some states distribute pulses and edible oils. Public health programs need to emphasize dietary diversity in addition to the more traditional concerns of undernutrition. The magnitude of our results suggests that this long-term consequence of the Green Revolution is not trivial. A back-of-the-envelope calculation implies that the number of years of life lost due to diabetes is of similar magnitude to the years of life gained from the reduction in infant mortality documented in Bharadwaj et al. (2020).

India is not alone in facing a dual burden of malnutrition and diet-related chronic diseases. China and Mexico are two other prominent examples (Popkin 2001). A dramatic rise in obesity and diabetes in Mexico has led to speculation that free trade agreements like the North American Free Trade Agreement (NAFTA) have

[34] Also related is the possibility that the composition of mothers who had children after the Green Revolution changed. Bharadwaj et al. (2020) show that the characteristics of women who gave birth were unaffected by the Green Revolution, suggesting that there is limited selection on this front.

contributed due to increased consumption of calorie-rich, low-nutrient food.[35] We leave an analysis of the causal impact of NAFTA on health to future work, but one policy implication of our paper is to highlight that policies that impact food production and distribution need to take into account changes in diet that may have adverse long-term consequences for health.

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